Name

Mathematics Stage 5 Paths

Algebraic Fractions with Algebraic Denominators

Book 1

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Contents

Syllabus Content 2
Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-ALG-P-01 simplifies algebraic fractions involving indices, and expands and factorises algebraic expressions

Apply the 4 operations involving algebraic fractions with pronumerals in the denominator

- Simplify algebraic fractions, including those involving indices
- Simplify expressions that involve operations with algebraic fractions, including algebraic fractions that involve pronumerals in the denominator and/or indices

Simplifying Algebraic Fractions involving Indices

ⓘ Producing Equivalent Algebraic Fractions

Produce equivalent fractions by multiplying the numerator and denominator by the same term.

Example Produce equivalent algebraic fractions

$$\frac{3n}{7} = \frac{\quad}{14}$$
$$= \frac{3n}{7} \frac{\times 2}{\times 2}$$
$$= \frac{6n}{14}$$

$$\frac{3a}{4b^2} = \frac{\quad}{12b^7}$$
$$= \frac{3a}{4b^2} \frac{\times 3b^5}{\times 3b^5}$$
$$= \frac{9ab^5}{12b^7}$$

Guided Practice

Produce equivalent fractions.

a

$$\frac{2y}{5} = \frac{\quad}{15}$$

$6y$

b

$$\frac{5a}{2} = \frac{\quad}{12}$$

$30a$

c

$$\frac{3t}{4w} = \frac{\quad}{16w}$$

$12t$

d

$$\frac{6x}{5y} = \frac{\quad}{5yz}$$

$6xz$

e

$$\frac{7m}{9n} = \frac{\quad}{18np}$$

$14mp$

f

$$\frac{5a}{4bc} = \frac{\quad}{8b^2c^3}$$

$10abc^2$

Simplifying Fractions involving Indices

- To simplify fractions:

1. Divide the numerator and denominator by a common factor.

- When indices are involved, use the index law:

$$\frac{a^m}{a^n} = a^{m-n}$$

Example Simplify algebraic fractions involving indices

$$\frac{10a^5}{5a^2}$$

$$= \frac{10a^5}{5a^2} \begin{matrix} \div a^2 \\ \div a^2 \end{matrix}$$

$$= \frac{10a^3}{5} \begin{matrix} \div 5 \\ \div 5 \end{matrix}$$

$$= 2a^3$$

$$\frac{8ab}{2a^2}$$

$$= \frac{8ab}{2a^2} \begin{matrix} \div a \\ \div a \end{matrix}$$

$$= \frac{8b}{2a} \begin{matrix} \div 2 \\ \div 2 \end{matrix}$$

$$= \frac{4b}{a}$$

Guided Practice

Simplify these expressions.

a	$\frac{35x^2}{7x}$	$5x$	**b**	$\frac{-14x^3}{7x}$	$-2x^2$
c	$-\frac{36ab^2}{4ab}$	$-9b$	**d**	$\frac{8xy^3}{4x^2y}$	$\frac{2y^2}{x}$
e	$\frac{-15pq^2}{30p^2q^2}$	$-\frac{1}{2p}$	**f**	$\frac{-20s}{45s^2t}$	$-\frac{4}{9st}$

Key Ideas

1. Simplify fractions by dividing the …… and …… by the …… ……

1 Write the equivalent fraction.

a $\frac{4m}{3} = \frac{}{6}$ $8m$

b $\frac{5m}{2n} = \frac{}{6n}$ $15m$

c $\frac{8}{5q} = \frac{}{30q}$ 48

d $\frac{3a}{4b} = \frac{}{4b^2}$ $3ab$

e $\frac{2u}{3w} = \frac{}{15wx}$ $10ux$

f $\frac{7w}{6x} = \frac{}{18x^2}$ $21wx$

2 Simplify the following:

a $\frac{x}{x}$ 1

b $\frac{a}{a^2}$ $\frac{1}{a}$

e $\frac{2x}{x}$ 2

f $\frac{3x^2}{9xy}$ $\frac{x}{3y}$

j $\frac{x}{2x}$ $\frac{1}{2}$

k $\frac{12ab}{4a^2b}$ $\frac{3}{a}$

3 Simplify the following:

a $\frac{3}{12}$ $\frac{1}{4}$

b $\frac{3x}{12}$ $\frac{x}{4}$

c $\frac{3}{12x}$ $\frac{1}{4x}$

d $\frac{12x}{3}$ $4x$

e $\frac{12x}{3x}$ 4

f $\frac{12x}{6x}$ 2

g $\frac{12xy}{6x}$ $2y$

h $\frac{6x}{12xy}$ $\frac{1}{2y}$

i $\frac{10x}{12xy}$ $\frac{5}{6y}$

j $\frac{10x}{12xy^2}$ $\frac{5}{6y^2}$

k $\frac{10xy}{12xy^2}$ $\frac{5}{6y}$

l $\frac{10x^2y}{12xy^2}$ $\frac{5x}{6y}$

m $\frac{10x^2y^2}{12xy^2}$ $\frac{5x}{6}$

n $\frac{10x^2y^3}{12xy^2}$ $\frac{5xy}{6}$

o $\frac{100x^2y^4}{120xy^2}$ $\frac{5xy^2}{6}$

4 Simplify the following:

a $\frac{35x^2}{7x}$ $5x$	**b** $-\frac{14x^2y}{7xy}$ $-2x$	**c** $\frac{-36ab^2}{4ab}$ $-9b$
d $\frac{8xy^3}{-4xy^2}$ $-2y$	**e** $\frac{-15pq^2}{-30p^2q^2}$ $\frac{1}{2p}$	**f** $-\frac{20s}{45s^2t}$ $-\frac{4}{9st}$

5 Simplify by writing as a fraction first.

a $12ab \div 2$ $6ab$	**b** $-10 \div 2gh$ $-\frac{5}{gh}$	**c** $8x \div x$ 8
d $-3xy \div yx$ -3	**e** $7mn \div 3m$ $\frac{7n}{3}$	**f** $-27pq \div 6p$ $-\frac{9q}{2}$
g $24ab^2 \div 8ab$ $3b$	**h** $25x^2y \div (-5xy)$ $-5x$	**i** $9m^2n \div 18mn$ $\frac{m}{2}$

6 Simplify the following expressions.

a $\frac{a^4}{a}$ a^3	**b** $\frac{3b^3}{9b}$ $\frac{b^2}{3}$	**c** $\frac{4ab^3}{12ab}$ $\frac{b^2}{3}$	**d** $\frac{6a^4b^2}{16a^3b}$ $\frac{3ab}{8}$
e $-\frac{2a^5}{3a^2}$ $-\frac{2a^3}{3}$	**f** $-\frac{8a^5}{20a^7}$ $-\frac{2}{5a^2}$	**g** $\frac{4a^3}{10a^8}$ $\frac{2}{5a^5}$	**h** $\frac{3a^3b}{12ab^4}$ $\frac{a^2}{4b^3}$
i $\frac{15a^4b^2}{5ab}$ $3a^3b$	**j** $\frac{28a^3b^5}{7a^4b^2}$ $\frac{4b^3}{a}$	**k** $\frac{2a^5b^2}{6a^2b^3}$ $\frac{a^3}{3b}$	**l** $-\frac{5a^3b^7}{10a^2b^{10}}$ $-\frac{a}{2b^3}$

7 Simplify the following expressions. Remember to follow the order of operations.

a $\dfrac{2ab \times 3bc \times 4cd}{4a \times 3bc \times 2d}$

bc

b $\dfrac{12a^2b + 4a^2b}{4b+2b}$

$\frac{8a^2}{3}$

c $\dfrac{7x^2y - 5yx^2}{12xy}$

$\frac{x}{6}$

d $\dfrac{8a^2b + (4a \times 2ba)}{3ba - 2ba}$

$16a$

e $\dfrac{10abc + 5cba + 5a \times bc}{4c \times 10ab}$

$\frac{1}{2}$

f $\dfrac{10x^2y - (4x \times 6xy)}{7xy^2}$

$\frac{-2x}{y}$

Adding and Subtracting Algebraic Fractions

Adding and Subtracting Algebraic Fractions

1. Rewrite the fractions with the lowest common multiple of the denominators.
2. Add or subtract the numerators.

Example Add/subtract algebraic fractions with pronumeral denominators

$$\frac{2}{5}+\frac{3}{a^2}$$

1. Same denominators: $\frac{\quad}{5a^2}+\frac{\quad}{5a^2}$

$$\frac{2a^2}{5a^2}+\frac{15}{5a^2}$$

2. Add numerators: $\frac{2a^2+15}{5a^2}$

$$\frac{2}{3x}-\frac{3}{5x}$$

1. Same denominators: $\frac{\quad}{15x}-\frac{\quad}{15x}$

$$\frac{10}{15x}-\frac{9}{15x}$$

2. Subtract numerators: $\frac{1}{15x}$

Guided Practice

Simplify these expressions.

a $\frac{1}{a}+\frac{1}{3a}$

$=\frac{\quad}{3a}+\frac{\quad}{3a}$

$=\frac{\quad}{3a}$

$\frac{4}{3a}$

b $\frac{1}{2x}-\frac{1}{3x}$

$=\frac{\quad}{6x}-\frac{\quad}{6x}$

$=$

$\frac{1}{6x}$

c $\frac{2}{a}-\frac{3}{4a}$

$\frac{5}{4a}$

d $\frac{3}{5x}-\frac{1}{2x}$

$\frac{1}{10x}$

e $\frac{5}{a}+\frac{2}{a^3}$

$\frac{5a^2+2}{a^3}$

f $\frac{1}{x^2}-\frac{1}{y}$

$\frac{y-x^2}{x^2y}$

Key Ideas

1. To add and subtract two fractions, the …… must be the same.
2. First, we find the …… …… …… of the denominators.
3. To change the denominator to the common multiple, we …… the numerator and denominator by the same ……
4. Once the denominators are the same, we …… or …… the numerators.

1 Add and subtract these fractions.

a i $\frac{3}{5}+\frac{1}{10}$ $\frac{7}{10}$	**ii** $\frac{3}{40}+\frac{1}{20}$ $\frac{1}{8}$	**iii** $\frac{3}{z}+\frac{1}{2z}$ $\frac{7}{2z}$	**iv** $\frac{3}{2x}+\frac{1}{x}$ $\frac{5}{2x}$
b i $\frac{3}{5}-\frac{1}{10}$ $\frac{1}{2}$	**ii** $\frac{3}{40}-\frac{1}{20}$ $\frac{1}{40}$	**iii** $\frac{3}{z}-\frac{1}{2z}$ $\frac{5}{2z}$	**iv** $\frac{3}{2x}-\frac{1}{x}$ $\frac{1}{2x}$

2 Simplify the following.

a $\frac{3}{x}+\frac{5}{2x}$ $\frac{11}{2x}$	**b** $\frac{7}{3x}-\frac{2}{x}$ $\frac{1}{3x}$	**c** $\frac{7}{4x}-\frac{5}{2x}$ $-\frac{3}{4x}$
d $\frac{4}{3x}+\frac{2}{9x}$ $\frac{14}{9x}$	**e** $\frac{3}{4x}-\frac{2}{5x}$ $\frac{7}{20x}$	**f** $\frac{2}{3x}+\frac{1}{5x}$ $\frac{13}{15x}$
g $\frac{-3}{4x}-\frac{7}{x}$ $-\frac{31}{4x}$	**h** $\frac{-5}{3x}-\frac{3}{4x}$ $-\frac{29}{12x}$	**i** $\frac{5}{3x}+\frac{3}{4x}$ $\frac{29}{12x}$
j $\frac{17}{100x}-\frac{7}{50x}$ $\frac{3}{100x}$	**k** $\frac{113}{4y}+\frac{1}{2y}-\frac{3}{y}$ $\frac{103}{4y}$	**l** $\frac{97}{6k}-\frac{5}{3k}+\frac{1}{k}$ $\frac{31}{2k}$

3 Simplify the following.

a $\frac{3}{x}+\frac{2}{x^2}$ $\frac{3x+2}{x^2}$	**b** $\frac{5}{x^2}+\frac{4}{x}$ $\frac{5+4x}{x^2}$	**c** $\frac{7}{x}+\frac{3}{x^2}$ $\frac{7x+3}{x^2}$
d $\frac{4}{x}-\frac{5}{x^2}$ $\frac{4x-5}{x^2}$	**e** $\frac{3}{x^2}-\frac{8}{x}$ $\frac{3-8x}{x^2}$	**f** $-\frac{4}{x^2}+\frac{1}{x}$ $\frac{x-4}{x^2}$
g $\frac{3}{x}-\frac{7}{2x^2}$ $\frac{6x-7}{2x^2}$	**h** $\frac{-2}{3x}+\frac{3}{x^2}$ $\frac{-2x+9}{3x^2}$	**i** $-\frac{5}{x^2}-\frac{3}{4x}$ $\frac{-20-3x}{4x^2}$

4 Simplify the following. You will need to rewrite the fractions with a common denominator.

a $\frac{2}{x}+\frac{x}{4}$

$\frac{8+x^2}{4x}$

b $\frac{-5}{x}+\frac{x}{2}$

$\frac{-10+x^2}{2x}$

c $\frac{-2}{x}-\frac{4x}{3}$

$\frac{-6-4x^2}{3x}$

d $\frac{3}{2x}-\frac{5x}{4}$

$\frac{6-5x^2}{4x}$

e $\frac{3x}{4}-\frac{5}{6x}$

$\frac{9x^2-10}{12x}$

f $\frac{1}{3x}-\frac{x}{9}$

$\frac{3-x^2}{9x}$

g $-\frac{2}{5x}+\frac{3x}{2}$

$\frac{-4+15x^2}{10x}$

h $\frac{-5}{4x}-\frac{3x}{10}$

$\frac{-25-6x^2}{20x}$

i $-\frac{6x}{4}-\frac{5}{3x}$

$\frac{-9x^2-10}{6x}$

5 Simplify these expressions.

a $\frac{3x}{2}+\frac{4x}{5}$

$\frac{23x}{10}$

b $\frac{x}{2}+\frac{4x}{5}$

$\frac{13x}{10}$

c $\frac{x}{2}+\frac{x}{5}$

$\frac{7x}{10}$

d $\frac{x}{2}-\frac{x}{5}$

$\frac{3x}{10}$

e $\frac{x}{5}-\frac{x}{2}$

$\frac{-3x}{10}$

f $\frac{5x}{5}-\frac{x}{2}$

$\frac{x}{2}$

g $\frac{5}{x}+\frac{2}{x}$

$\frac{7}{x}$

h $\frac{5}{x^2}+\frac{2}{x}$

$\frac{5+2x}{x^2}$

i $\frac{5}{2x^2}-\frac{2}{x}$

$\frac{5-4x}{2x^2}$

j $\frac{6}{2x^2}+\frac{2}{x}$

$\frac{3+2x}{x^2}$

k $\frac{6}{2xy}+\frac{2}{x}$

$\frac{3+2y}{xy}$

l $\frac{6}{2xy}+\frac{2}{3x}$

$\frac{9+2y}{3xy}$

m $\frac{6}{2xy}-\frac{2}{3x^2}$

$\frac{9x-2y}{3x^2y}$

n $\frac{6y}{2xy}+\frac{2}{3x^2}$

$\frac{9x+2}{3x^2}$

o $\frac{y}{2xy}-\frac{2}{3x^3}$

$\frac{3x^2-4}{6x^3}$

6 Simplify these expressions.

a $\frac{2}{3x} + \frac{5}{x} - \frac{1}{x}$

$\frac{14}{3x}$

b $-\frac{1}{2x} + \frac{2}{x} - \frac{4}{3x}$

$\frac{1}{6x}$

c $-\frac{4}{5x} - \frac{1}{2x} + \frac{3}{4x}$

$-\frac{11}{20x}$

d $\frac{4}{x^2} + \frac{3}{2x} - \frac{5}{3x}$

$\frac{24 - x}{6x^2}$

e $\frac{5}{x} - \frac{3}{2x^2} - \frac{5}{7x}$

$\frac{60x - 21}{14x^2}$

f $\frac{2}{x^2} - \frac{4}{9x} - \frac{5}{3x^2}$

$\frac{3 - 4x}{9x^2}$

g $\frac{2}{x} + \frac{x}{5} - \frac{x}{3}$

$\frac{30 - 2x^2}{15x}$

h $\frac{3x}{2} - \frac{1}{2x} + \frac{x}{3}$

$\frac{11x^2 - 3}{6x}$

i $-\frac{4x}{9} + \frac{2}{5x} + \frac{2x}{5}$

$\frac{18 - 2x^2}{45x}$

Multiplying and Dividing Algebraic Fractions

Multiplying Algebraic Fractions

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

Example Add/subtract algebraic fractions with pronumeral denominators

$$\frac{2a}{5} \times \frac{3b}{7}$$
$$= \frac{2a \times 3b}{5 \times 7}$$
$$= \frac{6ab}{35}$$

$$\frac{4x}{15} \times \frac{3}{2x}$$
$$= \frac{4x \times 3}{15 \times 2x}$$
$$= \frac{12x}{30x} = \frac{2}{5}$$

Guided Practice

Simplify these expressions.

a $\frac{x}{3} \times \frac{2}{5}$ $\frac{2x}{15}$

b $\frac{2}{3} \times \frac{4a}{5}$ $\frac{8a}{15}$

c $\frac{6x}{5} \times \frac{7y}{6}$ $\frac{7xy}{5}$

d $\frac{8a}{5} \times \frac{3b}{4c}$ $\frac{6ab}{5c}$

e $\frac{3x}{2} \times 6$ $9x$

f $\frac{10x}{3} \times \frac{3}{10x}$ 1

Key Ideas

1. To multiply two fractions, we multiply the ..... and .....

Dividing Algebraic Fractions

$$\frac{a}{b} \div \frac{c}{d}$$
$$= \frac{a}{b} \times \frac{d}{c}$$

Example Add/subtract algebraic fractions with pronumeral denominators

$$\frac{3a}{8} \div \frac{b}{5}$$
$$= \frac{3a}{8} \times \frac{5}{b}$$
$$= \frac{15a}{8b}$$

$$\frac{u}{4} \div \frac{15p}{2}$$
$$= \frac{u}{4} \times \frac{2}{15p}$$
$$= \frac{2u}{60p}$$
$$= \frac{u}{30p}$$

Guided Practice

Simplify these expressions.

a $\frac{3a}{5} \div \frac{1}{5}$

$3a$

b $\frac{9a}{10} \div \frac{1}{4}$

$\frac{18a}{5}$

c $\frac{4}{5} \div \frac{2y}{3}$

$\frac{6}{5y}$

d $\frac{4a}{7} \div \frac{2}{5}$

$\frac{10a}{7}$

e $\frac{4x}{5} \div 3$

$\frac{4x}{15}$

f $2 \div \frac{x}{5}$

$\frac{10}{x}$

Key Ideas

1. Dividing by a fraction is the same as multiplying by the ……

1 Simplify these products. Remember to simplify before you multiply.

a $\frac{6x}{5} \times \frac{7y}{6}$

$\frac{7xy}{5}$

b $\frac{2b}{5} \times \frac{7d}{6}$

$\frac{7bd}{15}$

c $\frac{8a}{5} \times \frac{3b}{4c}$

$\frac{6ab}{5c}$

d $\frac{9d}{2} \times \frac{4e}{7}$

$\frac{18de}{7}$

e $\frac{3x}{2} \times \frac{1}{6x}$

$\frac{1}{4}$

f $\frac{4}{9k} \times \frac{3k}{2}$

$\frac{2}{3}$

2 Simplify these divisions. Remember to simplify before you multiply.

a $\frac{3a}{4} \div \frac{1}{5}$

$= \frac{3a}{4} \times \frac{5}{1} =$

$\frac{15a}{4}$

b $\frac{2x}{5} \div \frac{3}{7}$

$\frac{4x}{15}$

c $\frac{9a}{10} \div \frac{1}{4}$

$\frac{18a}{5}$

d $\frac{2}{3} \div \frac{4x}{7}$

$\frac{7}{6x}$

e $\frac{4}{5} \div \frac{2y}{3}$

$\frac{6}{5y}$

f $\frac{1}{7} \div \frac{2}{x}$

$\frac{x}{14}$

g $\frac{4a}{7} \div \frac{2}{5}$

$\frac{10a}{7}$

h $\frac{4b}{7} \div \frac{2c}{5}$

$\frac{10b}{7c}$

i $\frac{2x}{5} \div \frac{4y}{3}$

$\frac{3x}{10y}$

j $\frac{2y}{x} \div \frac{3}{y}$

$\frac{2y^2}{3x}$

k $\frac{5}{12x} \div \frac{7x}{2}$

$\frac{5}{42x^2}$

l $\frac{4a}{5} \div \frac{2b}{7a}$

$\frac{14a^2}{5b}$

3 Simplify the following.

a $3 \times \frac{4x}{5}$

$\frac{12x}{5}$

b $\frac{4x}{5} \div 3$

$\frac{4x}{15}$

c $2 \div \frac{x}{5}$

$\frac{10}{x}$

d $4 \times \frac{a}{3}$

$\frac{4a}{3}$

e $5 \times \frac{7}{10x}$

$\frac{7}{2x}$

f $1 \div \frac{x}{2}$

$\frac{2}{x}$

4 Simplify these expressions.

a $\frac{3}{5} \times \frac{x}{y}$

$\frac{3x}{5y}$

b $\frac{x}{y} \times \frac{3}{5}$

$\frac{3x}{5y}$

c $\frac{x}{y} \times \frac{3}{6}$

$\frac{x}{2y}$

d $\frac{x}{y} \times \frac{3}{6x}$

$\frac{1}{2y}$

e $\frac{x}{xy} \times \frac{3}{6}$

$\frac{1}{2y}$

f $\frac{7x}{xy} \times \frac{3}{6}$

$\frac{7}{2y}$

g $\frac{7x}{xy} \times \frac{3}{6y}$

$\frac{7}{2y^2}$

h $\frac{7x}{xy} \div \frac{3}{6y}$

14

i $\frac{7x}{xy} \div \frac{6y}{3}$

$\frac{7}{2y^2}$

j $\frac{7x}{xy} \times \frac{6y}{3}$

14

k $\frac{7x}{xy} \times \frac{6y}{3} \times \frac{4y}{14}$

$4y$

l $\frac{7x}{xy} \div \frac{6y}{3} \times \frac{4y}{14}$

$\frac{1}{y}$

l $\frac{7x}{xy} \div \frac{6y}{3} \div \frac{4y}{14}$

$\frac{49}{4y^3}$

m $\frac{7x^2}{xy} \div \frac{6y}{3} \div \frac{4y}{14}$

$\frac{49x}{4y^3}$

n $\frac{7x^2}{xy} \div \left(\frac{6y}{3} \div \frac{4y}{14}\right)$

$\frac{x}{y}$

o $\frac{7x^2}{xy} \div \left(\frac{6y}{3} \times \frac{4y}{14}\right)$

$\frac{49x}{4y^3}$

p $\frac{7x^2}{xy} \div \left(\frac{6y}{9} \times \frac{4y}{14}\right)$

$\frac{147x}{4y^3}$

q $\frac{7x^2}{xy} \times \left(\frac{6y}{9} \div \frac{4y}{14}\right)$

$\frac{49x}{3y}$

5 The width of a rectangle is $\frac{3x}{5}$ cm. Its length is four times its width.

Write an expression for the area of the rectangle.

$\frac{36x^2}{25}$

6 Simplify these expressions.

a $x \times \frac{1}{x}$

1

b $x \div \frac{1}{x}$

x^2

c $x \div 3$

$\frac{x}{3}$

d $\frac{a}{b} \div c$

$\frac{a}{bc}$

e $a \div \frac{b}{c}$

$\frac{ac}{b}$

f $x \div \frac{x}{y}$

y

7 **HSC Advanced Sample Question Band 3**

Find the reciprocal of $\frac{1}{a} + \frac{1}{b} - \frac{c}{ab}$

$\frac{ab}{a+b-c}$

8 **HSC Advanced Sample Question Band 4**

Worker A picks a bucket of blueberries in a hours.
Worker B picks a bucket of blueberries in b hours.

Write an algebraic expression for the fraction of a bucket of blueberries that could be picked in one hour if A and B worked together.

$\frac{a+b}{ab}$

What does the reciprocal of this fraction represent?

$\frac{a+b}{ab}$ represents blueberry buckets per hour

So, the reciprocal represents hours per bucket of blueberries

Algebraic Fractions with Indices

Algebraic Fractions with Indices

- When working with algebraic fractions with indices, it is helpful to write out every line of working out. Messy work leads to more mistakes.

Example Multiply and divide algebraic fractions involving indices

$$\frac{a^2b^4}{6} \times \frac{9}{a^2b^2}$$
$$= \frac{9a^2b^4}{6a^2b^2}$$
$$= \frac{9b^2}{6}$$
$$= \frac{3b^2}{2}$$

$$\frac{3x^2}{8y^5} \div \frac{15x^3}{4y}$$
$$= \frac{3x^2}{8y^5} \times \frac{4y}{15x^3}$$
$$= \frac{12x^2y}{120x^3y^5}$$
$$= \frac{12}{120xy^4}$$
$$= \frac{1}{10xy^4}$$

Guided Practice

Simplify these expressions.

a $\frac{x^4}{n^2} \times \frac{m}{n^3}$ $\frac{mx^4}{n^5}$

b $\frac{x}{y} \times \frac{y^2}{x}$ y

c $\frac{xz}{5} \div \frac{x}{10z}$ $2z^2$

d $\frac{2x}{5y} \times \frac{2y}{5x^2}$ $\frac{4}{25x}$

e $\frac{3}{4y^3x} \times \frac{3x^2}{y^2}$ $\frac{9x}{4y^5}$

f $\frac{3x^2}{5y^2} \div \frac{2x}{3y}$ $\frac{9x}{10y}$

1 Simplify.

a $\frac{5a^2}{3} \times \frac{6}{2a}$

$5a$

b $\frac{2x^5}{2z^3} \times \frac{7z}{6x^9}$

$\frac{7}{6x^4z^2}$

c $\frac{a^2b^4}{8} \times \frac{12}{a^2b^2}$

$\frac{3b^2}{2}$

d $\frac{4m^3}{3n^7} \div \frac{5m^8}{6n^2}$

$\frac{8}{5m^5n^5}$

e $\frac{3x^2}{8y^5} \div \frac{15x^3}{4y}$

$\frac{1}{10xy^4}$

f $\frac{p^5q^4}{6} \div \frac{p^2q^2}{9}$

$\frac{3p^3q^2}{2}$

2 Simplify.

a $\frac{12a}{3c^3} \times \frac{6a^4}{4c^4}$

$\frac{6a^5}{c^7}$

b $\frac{24m^7n^5}{5m^3n} \times \frac{5m^2n^4}{8mn^2}$

$3m^5n^6$

c $\frac{p^4q^3}{p^2q} \div \frac{p^3q^2}{p^6q^4}$

p^5q^4

d $\frac{8k^4m^5}{5km^3} \times \frac{15km}{4k}$

$6k^3m^3$

e $\frac{12x^7y^3}{5x^4y} \div \frac{8xy^4}{25x^2y^3}$

$\frac{15x^4y}{2}$

f $\frac{9m^5n^2}{12mn} \times \frac{4mn^3}{m^4n^2} \div \frac{2m^2n}{m^3n^2}$

$\frac{3m^2n^3}{2}$